

# High Performance Track (HPT) – Applied Generative AI

## Supervised Project Tracks

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### Project 1: AI Knowledge Assistant (RAG System)

#### Focus

Build an AI system that operates on **private/proprietary data**.

#### Core Features

- Document upload (PDF/Text)
- Vector search (Pinecone)
- Context-based answering
- Source attribution (grounded responses)

#### Example Use Cases

- Resume improvement assistant
- Study material assistant
- Company knowledge base

#### Learning Focus

RAG pipeline, embeddings, retrieval optimization

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### Project 2: AI Workflow Automation Agent

#### Focus

Build an AI agent capable of executing **multi-step tasks and automating workflows**.

#### Core Features

- Task understanding
- Tool/function calling
- API integration
- Multi-step execution logic
- Saved actions/history

#### Example Use Cases

- Email drafting and sending assistant
- Task planning and execution agent
- Meeting notes to action generator

#### Learning Focus

Agents, structured outputs, workflow automation

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### Project 3: Multimodal AI Assistant (Voice/Image)

#### Focus

Build an AI system that supports **interaction through voice and/or images**.

#### Core Features

- Speech-to-text input
- Text-to-speech output
- Image understanding
- Interactive response system

#### Example Use Cases

- Voice-based tutor
- Interview practice assistant
- Image-based analysis tool

#### Learning Focus

Multimodal AI, user interaction design

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### Project 4: AI Decision & Insight Engine

#### Focus

Build an AI system that **analyzes data and generates insights and recommendations**.

## Core Features

- Input: structured and unstructured data
- AI-based analysis
- Insight generation
- Recommendations
- Report generation

## Example Use Cases

- Business analytics assistant
- Performance analysis tool
- Study progress evaluator

## Learning Focus

AI reasoning, structured outputs, decision systems

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# Submission Requirements

Each category must submit:

## 1. Working AI Application

- Fully functional system aligned with the selected project category
- Core features implemented (RAG / Agent / Multimodal / Analysis as applicable)

## 2. GitHub Repository

- Clean and well-structured codebase
- Proper README including:
  - Project overview
  - Setup instructions
  - Technology stack
- Environment configuration file (.env)

## 3. Documentation

- Basic system architecture (diagram or description)
- AI pipeline description (input → processing → output)
- Key design decisions

## 4. Deployment (Preferred)

- Live hosted application
  - or
  - Recorded demo (if deployment is not feasible)
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# Presentation Requirements

Each category will deliver a **final demo-based presentation** covering:

- **Problem Statement**  
What problem the system addresses
  - **Solution Overview**  
How the AI system solves the problem
  - **System Workflow**  
High-level application flow
  - **AI Implementation**
    - Use of RAG / agents / multimodal features
    - How outputs are generated
  - **Live Demonstration**  
Functional walkthrough of the system
  - **Key Outcomes**  
Practical impact and usability
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# Evaluation Criteria

Projects will be assessed based on:

- **Functionality** – completeness and correctness
- **AI Implementation** – effective and appropriate use of AI techniques
- **System Design** – clarity and structure of the solution

- **Code Quality** – organization and GitHub practices
  - **Presentation** – clarity, confidence, and demonstration
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## Expected Outcome

Each category is expected to deliver a **deployment-ready AI product** demonstrating:

- Practical application of generative AI
- End-to-end system development
- Ability to explain and justify technical decisions